

Class 5: Data Vis with ggplot2

Ryan Ellis (PID:A17673864)

Table of contents

Background	1
Add some custom features	4
Gene Expression	6
Going further	7

Background

There are many graphics systems in R for making plots and figures. These include so-called “*base R*” *graphics* like the `plot()` function and add on packages like **ggplot2**.

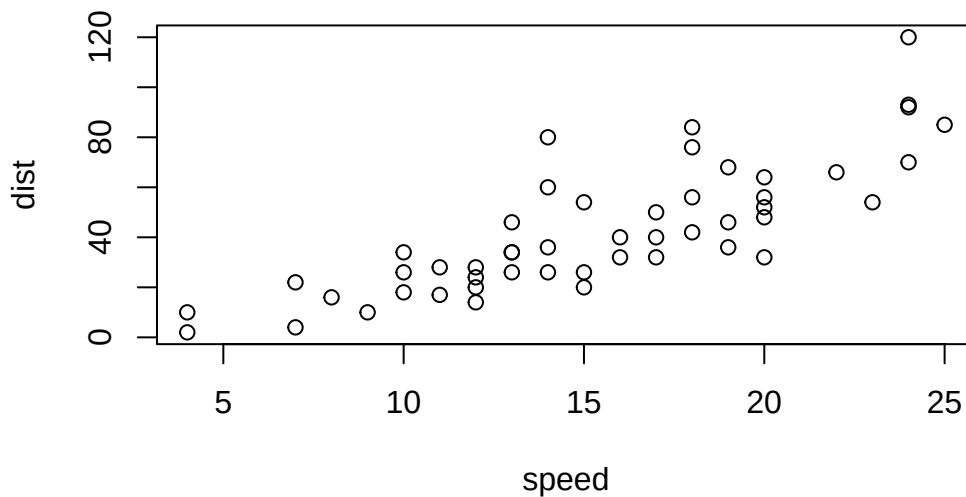
Let’s compare how we make a simple figure with these two systems:

We can use the in-built `cars` dataset:

```
head (cars)
```

```
  speed dist
1     4    2
2     4   10
3     7    4
4     7   22
5     8   16
6     9   10
```

```
plot(cars)
```



Before I can use ggplot 2 I need to install it on my computer. To do this we can use the function `install.packages("ggplot2")`

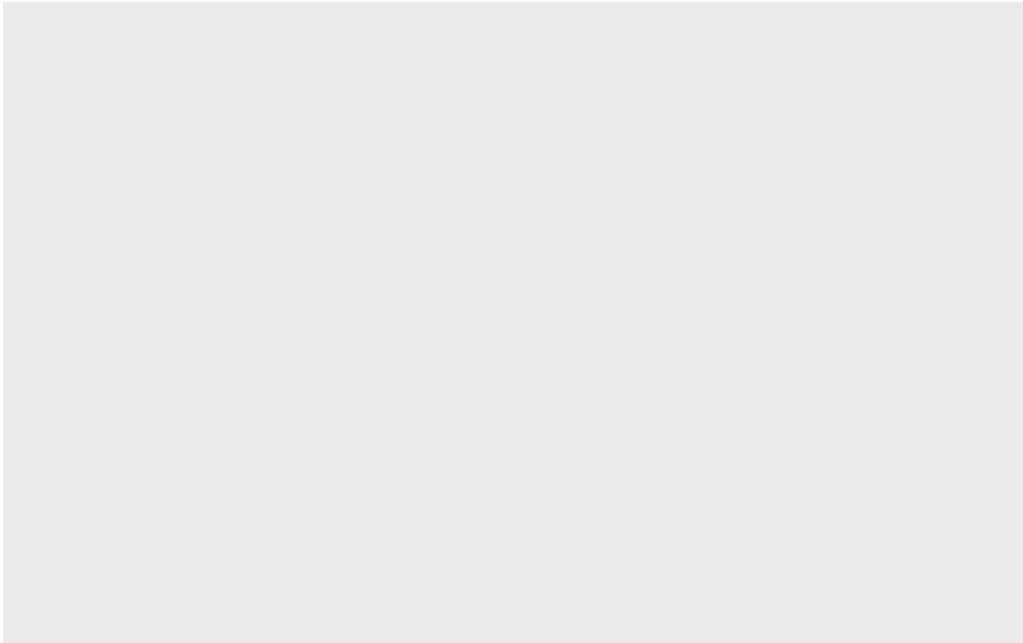
N.R We never run `install.packages()` in our quarto doc (we run it once only in the R console (R brain)), if we ran it in the doc it would reinstall the package everytime we render.

once installed we need to load the package into our R brain:

```
library(ggplot2)
```

The main function in **ggplot 2** package is called `ggplot()`

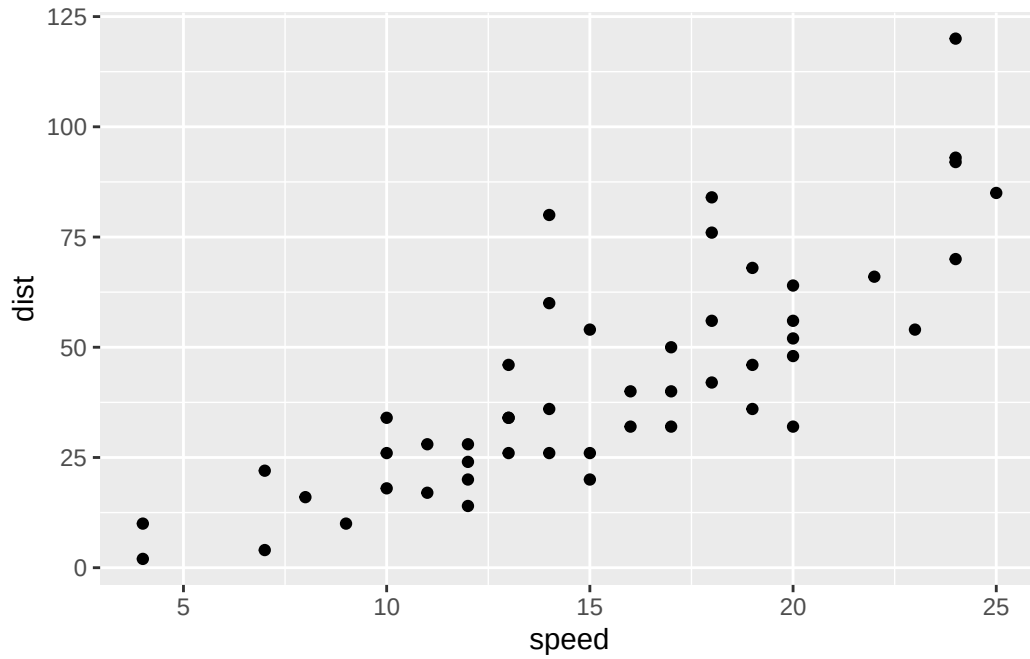
```
ggplot(cars)
```



Every ggplot has at least 3 layers:

- the **data** (a data.frame of the stuff we want to plot)
- the **aesthetics** (how the data maps to the plot)
- the **geom** layer (how you want the map to be drawn, ie. points, lines, etc.)

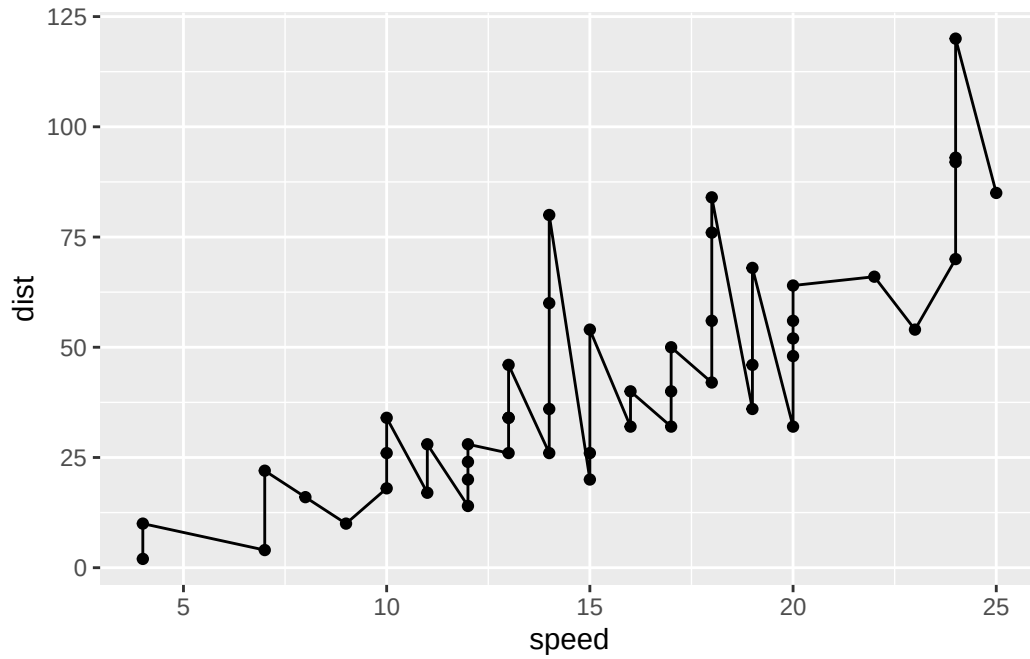
```
ggplot(cars)+  
  aes(x=speed , y=dist )+  
  geom_point()
```



Add some custom features

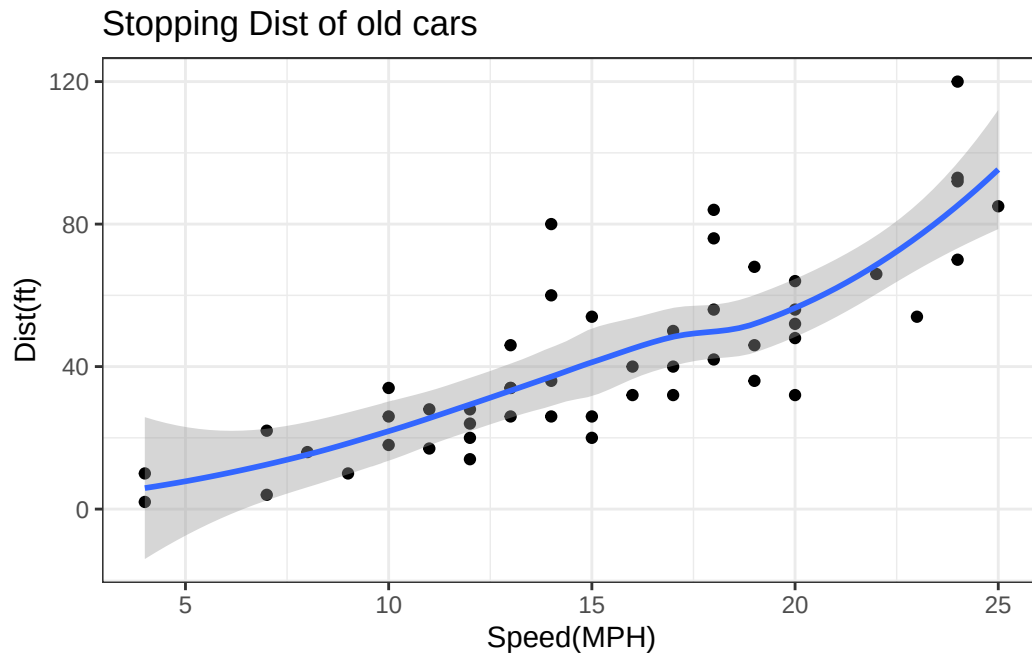
Let's add a trend line that shows a relationship between speed and distance

```
ggplot(cars)+  
  aes(x=speed , y=dist )+  
  geom_point()+  
  geom_line()
```



```
ggplot(cars)+  
  aes(x=speed , y=dist )+  
  geom_point()+  
  geom_smooth()+  
  theme_bw()+  
  labs(title= "Stopping Dist of old cars",  
        x="Speed(MPH)",  
        y="Dist(ft)")
```

`geom\_smooth()` using method = 'loess' and formula = 'y ~ x'



Q. Can you make `geom_smooth()` produce a linear straight line of best fit for the data and turn off the “grey” error region.

Gene Expression

```
url <- "https://bioboot.github.io/bimm143_S20/class-material/up_down_expression.txt"
genes <- read.delim(url)
head(genes)
```

	Gene	Condition1	Condition2	State
1	A4GNT	-3.6808610	-3.4401355	unchanging
2	AAAS	4.5479580	4.3864126	unchanging
3	AASDH	3.7190695	3.4787276	unchanging
4	AATF	5.0784720	5.0151916	unchanging
5	AATK	0.4711421	0.5598642	unchanging
6	AB015752.4	-3.6808610	-3.5921390	unchanging

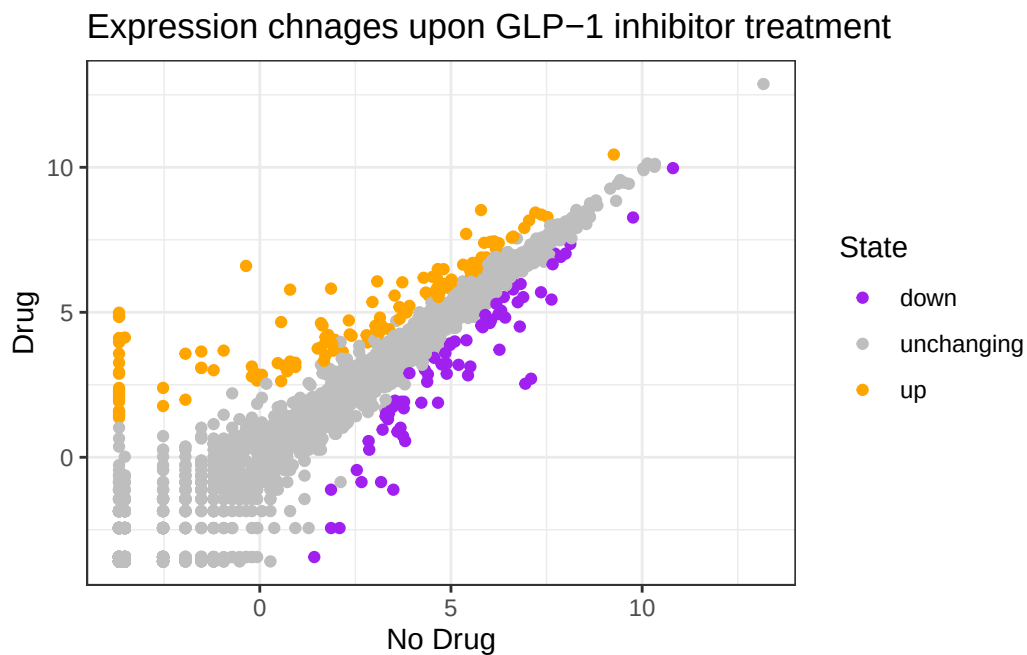
A useful new function in this context is the `table()` function

```
table(genes$state)
```

< table of extent 0 >

My first Plot attempt

```
ggplot(genes)+  
  aes(Condition1, Condition2, col=State)+  
  geom_point()+  
  scale_color_manual (values=c("purple", "gray", "orange"))+  
  theme_bw()+  
  labs(x= "No Drug", y= "Drug", title= "Expression chnages upon GLP-1 inhibitor treatment")
```



Going further

Here we read the famous gapminder data set:

```
# File location online  
url <- "https://raw.githubusercontent.com/jennybc/gapminder/master/inst/extdata/gapminder.tsv"
```

```
gapminder <- read.delim(url)
head(gapminder)
```

	country	continent	year	lifeExp	pop	gdpPercap
1	Afghanistan	Asia	1952	28.801	8425333	779.4453
2	Afghanistan	Asia	1957	30.332	9240934	820.8530
3	Afghanistan	Asia	1962	31.997	10267083	853.1007
4	Afghanistan	Asia	1967	34.020	11537966	836.1971
5	Afghanistan	Asia	1972	36.088	13079460	739.9811
6	Afghanistan	Asia	1977	38.438	14880372	786.1134

Q. How many entries (ie. rows) are there in the data set?

```
length(table(gapminder$country))
```

```
[1] 142
```

```
length(unique(gapminder$country))
```

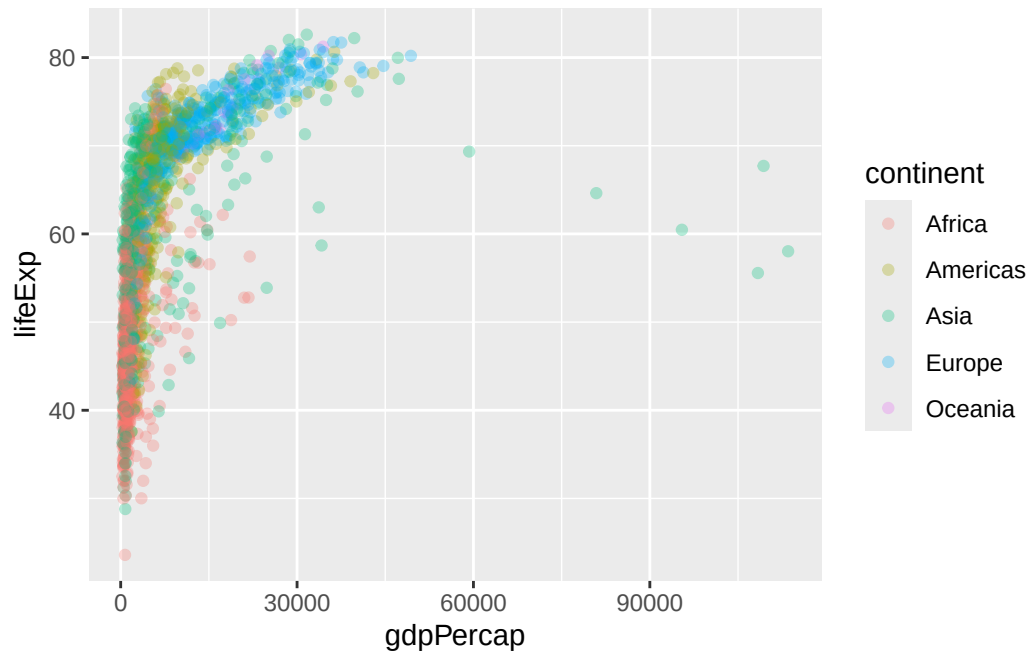
```
[1] 142
```

Lets make our first plot of the entire dataset:

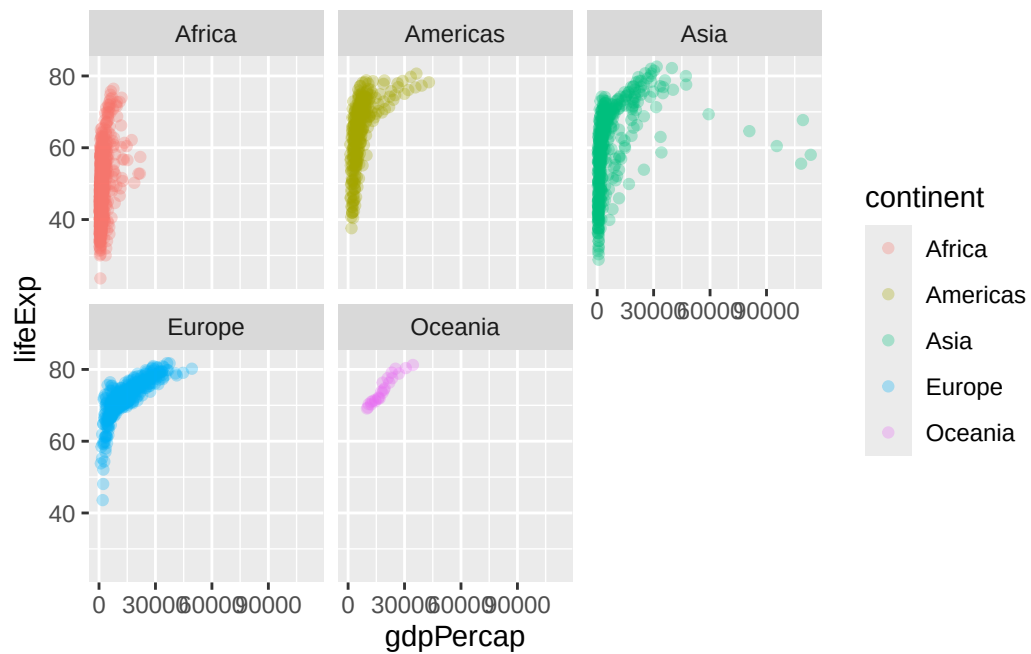
Plot of “gdpPercap” vs “lifeExp” colored by “continent”

```
p <- ggplot(gapminder)+
  aes(gdpPercap,lifeExp, color=continent)+
  geom_point(alpha=0.3)
```

```
p
```

```
p+
  facet_wrap(~continent)
```



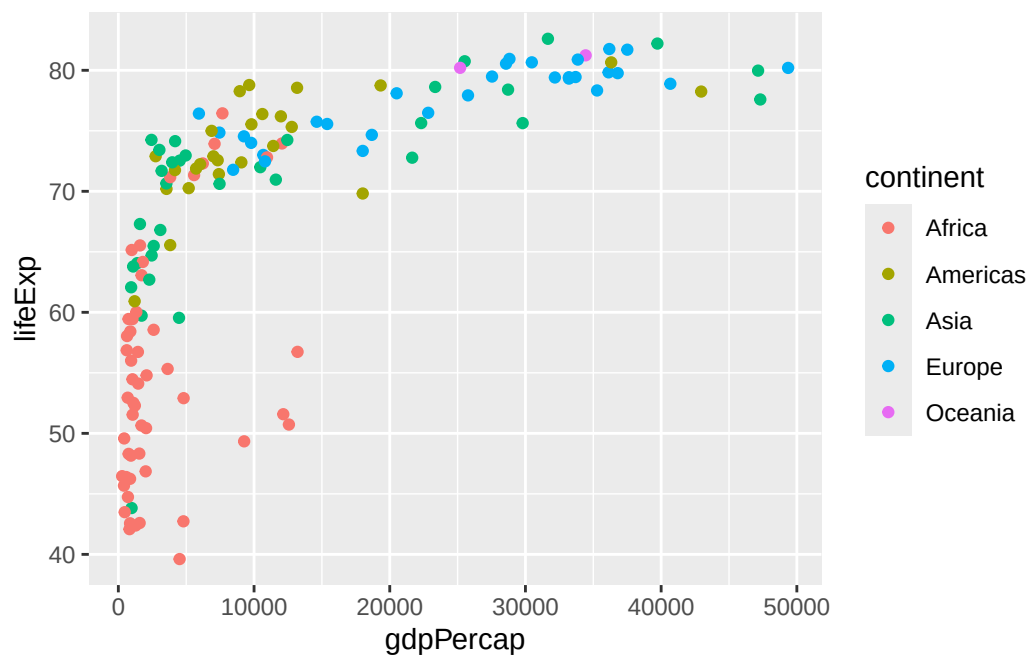
Make a plot for 1977 and 2007 only (not all years in the data set)

Q. First use the **dplyr** package and the `filter()` function from that package to extract the rows from the year 2007

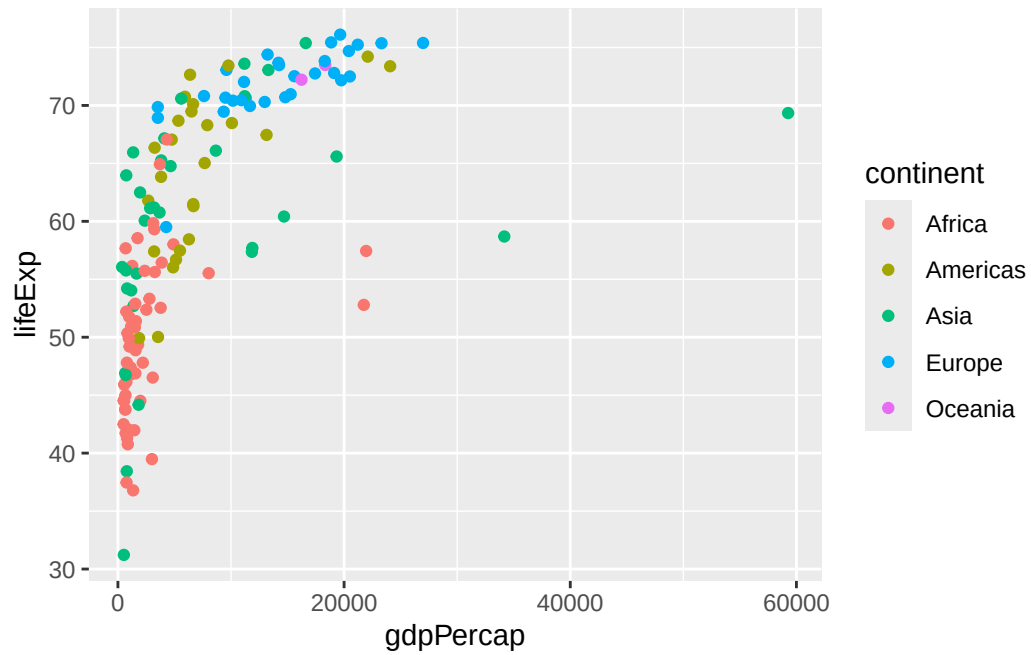
```
library(dplyr)
```

```
g07<- filter(gapminder, year==2007)
g77<- filter(gapminder, year==1977)
g<- filter(gapminder, year==1977 | year==2007)
```

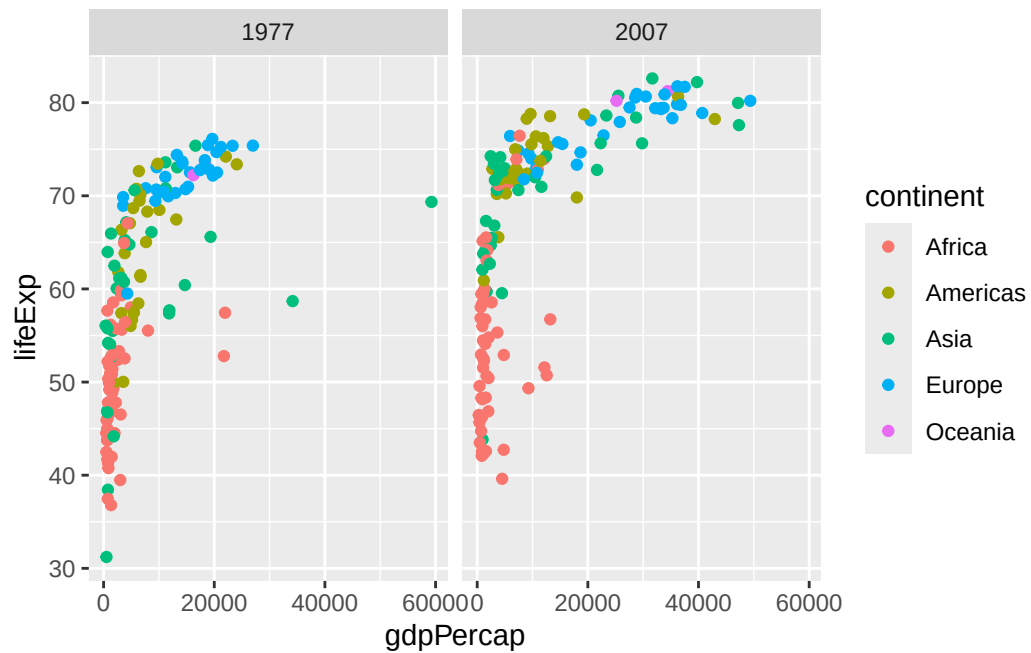
```
ggplot(g07)+
  aes(gdpPercap,lifeExp, color=continent)+
  geom_point()
```



```
ggplot(g77)+
  aes(gdpPercap,lifeExp, color=continent)+
  geom_point()
```

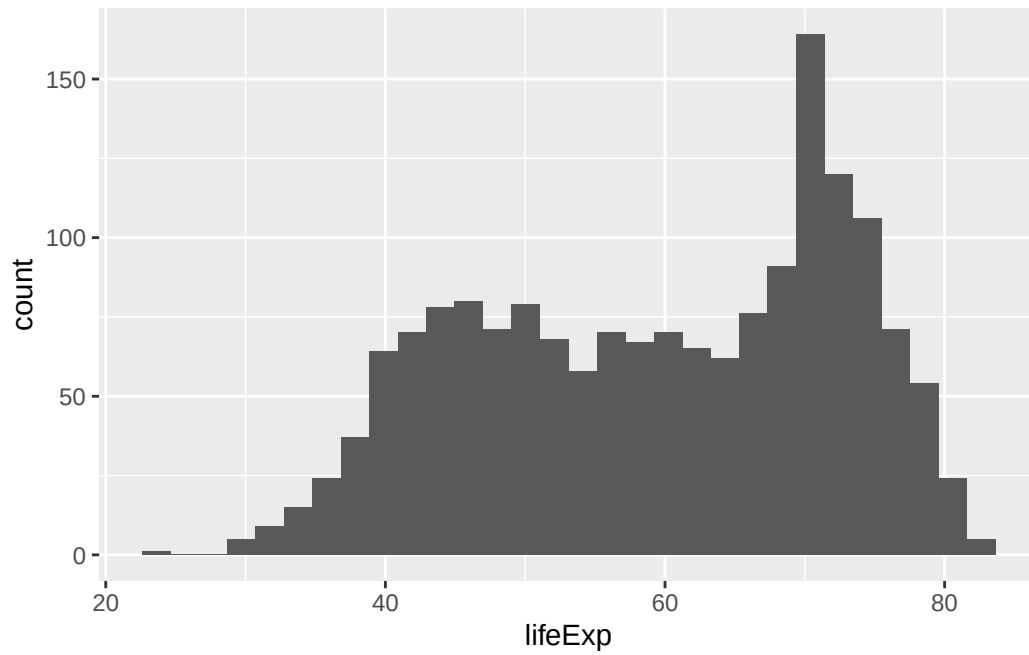


```
ggplot(g)+  
  aes(gdpPerCap,lifeExp, color=continent)+  
  geom_point()+  
  facet_wrap(~year)
```

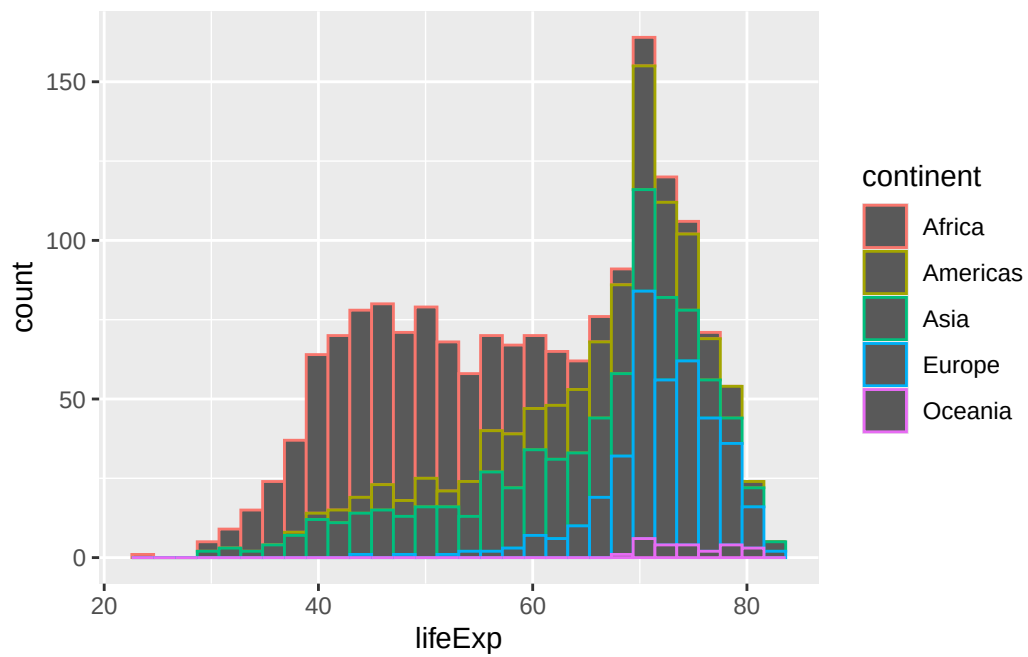


Q. Make a histogram of lifeExp faceted by continent Q. Make a histogram of lifeExp colored by continent (try using `fill=continent` or `col=continent`)

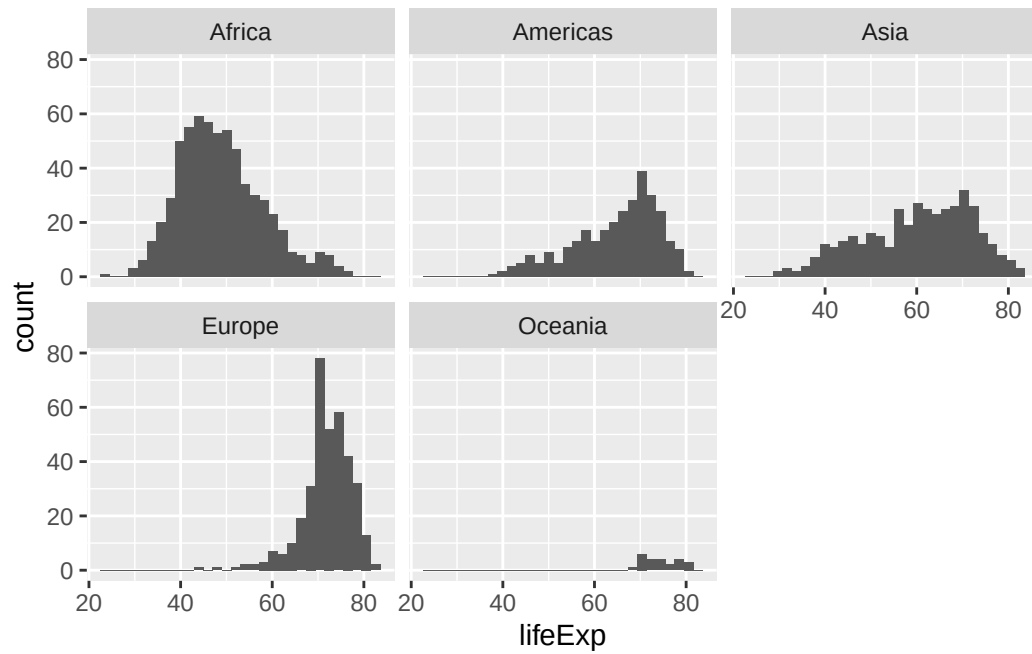
```
ggplot(gapminder)+
  aes(lifeExp)+
  geom_histogram()
```



```
ggplot(gapminder)+
  aes(lifeExp, col=continent)+
  geom_histogram()
```



```
ggplot(gapminder)+
  aes(lifeExp)+
  geom_histogram()+
  facet_wrap(~continent)
```



```
ggplot(gapminder)+
  aes(lifeExp, fill=continent)+
  geom_histogram()
```

